# Storm transposition

## CEVE 543

2026-10-26


## Summary

An event set is a long list of plausible storms with a probability attached to each. In this session we compute risk measures from an event set, build one for a watershed by moving observed storms around its region (stochastic storm transposition), weight storms that come with return periods, and see what a single rainfall number per storm leaves out.

- Average annual loss $\mathrm{AAL} = E[L]$, estimated by $\frac{1}{n}\sum_{i=1}^{n} L_i$ over $n$ equally likely synthetic years, where $L_i$ is the total loss in synthetic year $i$.
- $\mathrm{VaR}_\alpha$ is the annual loss exceeded with probability $1-\alpha$; TVaR, also called CVaR or expected shortfall, is $E[L \mid L \geq \mathrm{VaR}_\alpha]$.
- From a ladder of design events (the 2- to 100-year storms, each with its damage $d_k$), AAL is the area under damage against annual exceedance probability $p_k = 1/T_k$, approximated with trapezoids: $\mathrm{AAL} \approx \sum_k (p_k - p_{k+1})\frac{d_k + d_{k+1}}{2}$.
- From a set of storms with no synthetic years, fit the distribution of an intensity measure, sort the storms from the rarest, $T_1 > T_2 > \cdots$, and weight each by its own rate $\lambda_j = 1/T_j - 1/T_{j-1}$ with $1/T_0 = 0$; then $\mathrm{AAL} = \sum_j \lambda_j d_j$.
- An intensity measure is one number per storm, and its frequency need not match the frequency of flooding or damage.


For supplementary notes, read:

- C. O'Rourke and J. Burdick [1], pp. 9-10, on the exceedance probability curve, VaR, and TVaR.
- D. B. Wright, G. Yu, and J. F. England [2] §2-3, pp. 2-5, on the transposition procedure and its assumptions.
- J. A. Baer *et al.* [3], pp. 1-7, this week's reading.

## Bibliography


[1] C. O'Rourke and J. Burdick, "Catastrophe Modeling 101," White Paper, Jul. 2025. [Online]. Available: https://www.nb.com/

[2] D. B. Wright, G. Yu, and J. F. England, "Six Decades of Rainfall and Flood Frequency Analysis Using Stochastic Storm Transposition: Review, Progress, and Prospects," *Journal of Hydrology*, vol. 585, p. 124816, Jun. 2020, doi: 10.1016/j.jhydrol.2020.124816.

[3] J. A. Baer *et al.*, "Neglecting Spatiotemporal Rainfall Variability Misrepresents Flood Hazard and Risk," *npj Natural Hazards*, Jun. 2026, doi: 10.1038/s44304-026-00208-5.